Effects of Carbon Emission Taxes on Transportation Mode Selections and Social Welfare

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***Abstract*—In this paper, we analyze how carbon emissions affect the selection of transportation modes and social welfare by using a two-stage Stackelberg gaming model. Based on this model, the government's optimal carbon-emission tax scheme and the company's optimal transportation mode and production decisions are explored. We find that: 1) whether or not the transport carbon-emission tax can increase social welfare depends on the relationships among the social cost of carbon (SCC), the transportation mode shifting threshold (TMST), and the biggest carbon-emission tax that a company can afford (BCRA); 2) a greater SCC implies a higher probability of improving social welfare via imposing transportation carbon-emission tax; and 3) a smaller TMST or BCRA yields a higher probability of improving social welfare when a carbon-emission tax is imposed. Further study shows that imposing a carbon-emission tax on the product with a higher production cost, a bigger product volume, or a bigger product density can increase the probability of improving social welfare.**

***Index Terms*—Carbon-emission tax, economic systems, social welfare, supply chain management, sustainability, transportation mode selection.**

I. Introduction

GREEN supply chain management is a hot topic in industrial systems engineering [1]. Nowadays, supply chains are globalized and volumes of freight transportation are gradually increasing. According to the U.S. Department of Energy in 2005, the consumption of energy in freight transportation is approximately 6800 trillion British Thermal Unit (BTU), accounting for about 25.7% of the total energy used in nonmilitary transportation activities [8]. This consumption is forecasted to increase at a rate of 1.8% per year. As reported in [25], freight transport growth and energy consumption growth will lead to a similar increase in transport carbon emissions. Therefore, the increase in both freight transport and energy consumption, and the subsequent increase in carbon emissions, will put enormous pressure on the environment [9], [29], [30], [33], [36]. In order to solve this problem, it is necessary to reduce the carbon emissions associated with transportation. However, it will be difficult to reduce transport carbon emissions by only relying on market adjustment without any government intervention. If the companies are forced to reduce carbon emissions by regulations, the following questions should be addressed: 1) what kind of measures for reducing transportation carbon emissions should be encouraged and 2) should governments regulate the transport process of all types of products or just some types of products. This paper aims to provide answers to these important questions.

The total transportation carbon emissions are equal to the result of freight transport volume multiplied by carbon emission intensity (i.e., the emissions associated with transporting one unit of the product). Over the past few decades, companies have reduced transportation carbon emissions by either decreasing the amount of freight transportation or improving the transportation mode's carbon emission efficiency. In fact, companies can also improve their carbon emission efficiency by using more environmentally friendly transportation modes, such as trains and boats, under the intermodal transportation scheme [15]. However, intermodal transportation has many disadvantages. For example, each intermodal transshipment activity involves additional transaction and coordination costs, and there exists a high investment risk due to the sunk cost for intermodal transportation [2]. Therefore, governments generally provide some incentives (such as investment grants) [3] or "decentives" (such as the carbon-emission tax) to encourage the development of intermodal forms of transport. For instance, the German government offers incentives amounting up to 80% of the initial investment to companies if these companies construct intermodal terminals [2]. Because of this incentive, many companies have set up intermodal service terminals in Germany and used them in the market. This massive entry of small and medium service terminals provide more choices for freight transportation, but it also has a negative effect on efficiency [2]. On the other hand, the carbon-emission tax regimes and cap-and-trade mechanisms are usually considered as the efficient measures

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for reducing carbon emissions [4], [14], [21]. In fact, the carbon-emission tax can enhance the competitive advantage of clean transportation modes and it is easy to implement. However, the carbon emission tax will increase operating costs and product's selling price. Thus, the government needs to examine the tradeoff between benefit and loss in order to analyze whether or not a transport carbon-emission tax should really be implemented. Observe that the basic difference between the carbon-emission tax regime and the cap-and-trade mechanism [39] is that the carbon price in the carbon-emission tax regime is determined by the government while the carbon price in the cap-and-trade mechanism is determined by the carbon market, which is indirectly influenced by the government through the carbon emission permits. If the two carbon prices are equal, the two regimes will have the same effect on "social welfare."

Several areas of literature, including the sustainable transport strategies, the social cost of carbon (SCC), and the optimal transportation mode selection, are related to this research. We review them as follows. First, as sustainable transportation is becoming more and more important to our society, a lot of related studies were reported in recently years. For example, Ruamsook and Thomchick [27] summarized the existing research on environmental-friendly transportation strategies and constructed a framework to support the long-term and sustainable development of transportation systems. Methods for achieving sustainable transport development, from a macro perspective, in North America and Europe were examined in [4], [14], and [21]. However, these works did not analyze how carbon emission reduction schemes can be implemented in practice from a micro perspective. Liedtke and Murillo [2] assessed the welfare effects of two policies that could promote intermodal services, namely, the investment grants for terminal operators and the internalization of external cost (like the carbon-emission tax). They showed that in a high volume market, investment grants would have a negative effect on social welfare due to the decreased market efficiency. However, in the less developed markets, the investment grants would have a positive effect on social welfare. Mao *et al.* [18] explored and reported the effects of the carbon-emission tax, the energy tax, and the subsidies for cleaner vehicles on carbon reduction for the China transport sector. For other related studies, please refer to [28], [32], and [34].

To quantitatively analyze the cost and benefit of climate policies, the SCC concept was proposed in [16]. To be specific, SCC is an estimate of the present value of damages avoided from a marginal reduction in carbon emissions, and it represents the shadow price of carbon emissions [13], [35]. The value of SCC would have a significant effect on the intensity of carbon emissions reduction effort and the optimal policy's selection. Thus, estimating the value of SCC as accurately as possible is important and various models were proposed to help [12], [22]–[24], [26], [31]. In the modeling process, these three models simplify some complex factors of the natural, social, and economic systems. Other related studies on these models include [13], [19], and [35].

Another related field is the optimal transportation mode selection problem with environmental considerations. In this field, Meixell and Norbis [20] reviewed the existing research and found that very little prior studies had examined the factors in energy consumption or carbon emissions. Cholette and Venkat [7] calculated the emissions from transportation and warehousing perspectives and investigated the impact of different transportation modes. Several delivery options were also considered and evaluated in terms of costs and emissions. One of their findings is that the transportation mode has a large impact on the supply chain's total emissions. Using decision analysis methods including expert scoring, Leal and D'Agosto [17] considered the case of shipping bioethanol from a production location to its exportation port. They explored how to choose the optimal transportation mode while taking into account various socio-environmental factors. Hoen *et al.* [11] studied a shipper that has outsourced transport and decided to cap its carbon emissions from outbound logistics for a group of products. They focused on analyzing how transportation modes selection affects the shipper's profit without considering social welfare effects. For more related literature, refer to [5], [6], [10], [28], and [34].

Motivated by importance of the topic and related literature, we develop in this paper, a two-stage Stackelberg gaming model and investigate how the carbon-emission tax influences the equilibrium selling price, transportation mode selection, and transport carbon emissions. We further study the effects of carbon-emission taxation on social welfare and analyze the optimal transport carbon-emission taxation scheme. We find that the carbon-emission tax will push up the equilibrium selling price and have negative effects on both the firm's profits and the consumer surplus. In addition, we reveal that the carbon-emission tax can improve social welfare only when the value of SCC is bigger than at least one of the thresholds determined by the product characteristics (such as the market base, the sensitivity of sales to price, the production cost, the product density, and the volume) and the transportation mode characteristics (including the transportation cost and the emission efficiency). We also illustrate that the probability of improving social welfare via imposing the carbon-emission tax relies on the above mentioned thresholds.

The remainder of this paper is organized as follows. In Section II, we develop a two-stage Stackelberg model. In Section III, we present the equilibrium analysis. In Section IV, we present numerical examples. Section V concludes this paper.

II. Model

A. Model Description

In this paper, we develop a two-stage Stackelberg gaming model including two players, namely, the government and the "firm" (which represents an alliance unit of a company and a third party logistics (3PL) provider). In the first stage, the government imposes a transportation-based carbon-emission tax to improve social welfare. Note that social welfare includes several parts, including government revenues from the carbon-emission taxation scheme, social welfare losses due to carbon emissions, traditional welfare drivers of profits, and consumer surplus. The carbon-emission tax increases the government

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revenues and reduces the carbon emissions. However, at the same time, it increases the firm's operating costs. Therefore, the government needs to design an appropriate transportation carbon-emission tax to improve social welfare. Whether or not the carbon-emission tax can improve social welfare depends on a tradeoff between benefits and losses in which social welfare is involved in. In the second stage, the firm decides which transportation mode to use for the shipments and sets its selling price accordingly. In this paper, as we indicated above, the firm represents a shipper-carrier alliance which consists of a manufacturing company and a 3PL provider. To be specific, we consider a manufacturing company which produces one kind of product and outsources the transport function to a 3PL company. Under this scenario, shifting the transportation mode does not require a change in the supply chain. Traditionally, the optimal decision on transportation mode selection is based on the tradeoff between the transportation cost and the inventory related costs. We bring in the carbon-emission tax cost as a third component. Therefore, the transportation mode selection decision making problem considered in this paper is based on the tradeoff among the transport cost, the holding cost, and the carbon-emission tax. To derive more analytical results, the market demand is assumed to be equal to the product sales, and equal to the volume of freight transport. Without the carbon-emission tax, the alliance can get a positive profit no matter which transportation mode is used for the shipments.

Let $I = \{1, 2\}$ denote the set of available transportation modes (including intermodal transport). Note that unless a truck is used to execute the transport or both the origin and destination are located near the terminals, a combination of transportation modes (intermodal rail transport or intermodal water transport) is required. For the product shipped with transportation mode i, a transportation cost is incurred which is denoted by $c_i (c_i \geq 0)$, and is expressed as a monetary unit per ton of cargo shipped over 1 km (i.e., \$/ton-km). We denote the distance traveled for the product and the lead time by d_i (km) and l_i (days), respectively. They both depend on the transportation mode i being used.

To produce one unit of product, the company incurs a unit cost c^0(\$). The product volume is $v(\mathrm{m}^3)$, and the product density is $\rho(\mathrm{kg/m}^3)$. The amount of goods a vehicle can carry depends on the weight and the volume of the load. If an item is light yet large, not many items can be transported. Therefore, the transport cost is accounted based on the dimensional or volumetric weight of the shipment. The formula for weight is $w_i = v \max\{\bar{\rho}_i, \rho\}$, where $\bar{\rho}_i$ denotes the prespecified minimum density. Let u_i denote the transportation cost associated with using transportation mode i for one unit of the product. Then, u_i can be represented by $u_i = w_i d_i c_i$. We assume that the products are paid by the customer when they are delivered. Hence, one unit of the product in-transit corresponds to an opportunity cost of c^0. Let $h (h \geq 0)$ denote the opportunity cost (or the holding cost) for the items in-transit. Thus, the opportunity cost for one unit of the product in-transit is $\mathrm{hc}^0 l_i$. The emissions quantity associated with transporting one unit of the product (emissions intensity) with mode i is denoted by e_i. Based on the emission measurement methodology, we have $e_i = w_i(e_i^0 + d_i e_i^d)$, where e_i^0 and e_i^d are the mode-specific emission constants. The fixed emission factor $e_i^0 (e_i^0 \geq 0)$ is associated with the emissions generated during the beginning and end of a trip, and the variable emission factor $e_i^d (e_i^d > 0)$ denotes the emissions generated per kilometer traveled. In this paper, we assume $l_1 < l_2, c_1 > c_2$, and $e_1 > e_2$, where 1 represents "road transport" and 2 represents "rail transport." Let c^e(\$/kg) denote the carbon-emission tax, i.e., the carbon price. The carbon-emission tax associated with using transportation mode i for one unit of the product is $c^e e_i$. Let c^s(\$/kg) denote the value of SCC.

B. Model Development

We assume that the alliance sets the price and observes the demand. Let p_i denote the selling price of the product when transportation mode i is used and the demand (per time unit) is denoted by $q_i(p_i)$. We use an additive demand function $q_i(p_i) = Q - \epsilon p_i$, where $Q > 0$ corresponds to the maximum demand (market base) and $\epsilon > 0$ denotes the "price-demand sensitivity coefficient." To ensure that a nonnegative quantity is sold, we have $p_i \leq Q/\epsilon$.

The profit per time unit for transportation mode i, which is denoted by Π_i, is given as follows:

$$\Pi_i(p_i) = -\epsilon p_i^2 + \epsilon p_i \times \left(c^0 + u_i + \mathrm{hc}^0 l_i + c^e e_i + \frac{Q}{\epsilon}\right) - Q \times \left(c^0 + u_i + \mathrm{hc}^0 l_i + c^e e_i\right). \quad (1)$$

Let Γ_i denote the total carbon emissions associated with shipping the product using transportation mode i. Then, we have

$$\Gamma_i(p_i) = e_i \times (Q - \epsilon p_i). \quad (2)$$

Let CS_i denote the consumer surplus. It is given by

$$\mathrm{CS}_i(p_i) = \frac{1}{2}\left(\frac{Q}{\epsilon} - p_i\right)(Q - \epsilon p_i). \quad (3)$$

Let $\xi_i(p_i, c^s)$ denote the social welfare losses due to carbon emissions. Then, we have

$$\xi_i\left(p_i, c^s\right) = c^s e_i (Q - \epsilon p_i). \quad (4)$$

Further, the profit maximization model of the alliance is represented as follows:

$$\max_{i \in I} \max_{p_i} \{\Pi_i(p_i) | Q - \epsilon p_i \geq 0\} \quad (5)$$

and the government's social welfare maximization model is given below

$$\max_{c^e} \left\{W\left(c^e, p_k\left(c^e\right)\right) \middle| c^e \geq 0\right\} \quad (6)$$

where $W(c^e, p_k(c^e)) = \mathrm{CS}_k(p_k(c^e)) + \Pi_k(p_k(c^e)) + c^e \times \Gamma_k(p_k(c^e)) - \xi_k(p_k(c^e), c^s)$, and k and $p_k(c^e)$ are the optimal mode selection and the equilibrium selling price of the alliance, respectively, when the carbon-emission tax is c^e.

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III. Equilibrium Analysis

In this section, we solve the two-stage Stackelberg gaming model using backward induction and discuss the equilibrium results.

A. Equilibrium Selling Price

It is observed that $\Pi_i(p_i, c^e)$ is concave in $p_i(c^e)$. Let $p_i^*(c^e)$ maximize $\Pi_i(p_i, c^e)$, then

$$p_i^*(c^e) = \min\left\{\frac{1}{2}\left(c^0 + u_i + \mathrm{hc}^0 l_i + c^e e_i + \frac{Q}{\epsilon}\right), \frac{Q}{\epsilon}\right\} \quad (7)$$

$$q_i\left(p_i^*(c^e)\right) = \max\left\{\frac{1}{2}\left(Q - \epsilon \times \left(c^0 + u_i + \mathrm{hc}^0 l_i + c^e e_i\right)\right), 0\right\}. \quad (8)$$

From (7), it is obvious that the equilibrium selling price is increasing in the carbon-emission tax c^e. To ensure that a nonnegative quantity is sold, the biggest carbon-emission tax that the alliance can afford (BCRA) is given as follows:

$$r_i^e = \frac{1}{e_i} \times \left(\frac{Q}{\epsilon} - c^0 - u_i - \mathrm{hc}^0 l_i\right). \quad (9)$$

Denote the corresponding maximum profit by $\Pi_i^*(z_i(c^e)) = \Pi_i(p_i^*(c^e), z_i(c^e))$, we have

$$\Pi_i^*\left(z_i(c^e)\right) = \begin{cases} \frac{1}{4}\left(\epsilon \times \left(z_i(c^e) + c^0\right)^2 - 2Q \right. \\ \quad \left. \times \left(z_i(c^e) + c^0\right) + \frac{Q^2}{\epsilon}\right), & \text{if } 0 \le \mathrm{c^e} \le r_i^e \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where $z_i(c^e) = \tau_i + c^e e_i, \quad \tau_i = \mathrm{hc}^0 l_i + u_i$.

Proposition 1: Consider $i \in I$, the optimal profit function $\Pi_i^*(z_i(c^e))$ is decreasing and convex in $z_i(c^e)$. Furthermore, $\Pi_i^*(z_i(c^e))$ is decreasing in $c^e \in [0, r_i^e]$.

Proof: First, according to the second-order derivative $(d^2\Pi_i^*(z_i(c^e))/dz_i(c^e)^2) = (1/2)\epsilon > 0$, we show that $\Pi_i^*(z_i(c^e))$ is convex in $z_i(c^e)$, and $z_i(c^e) = (Q/\epsilon) - c^0$ minimizes the function $\Pi_i^*(z_i(c^e))$. Furthermore, $z_i(c^e) = \mathrm{hc}^0 l + u_i + e_i c^e$ is increasing in c^e, and r_i^e is the biggest value of c^e. So, we have $z_i(c^e) \le z_i(r_i^e) = Q/\epsilon - c^0$, which means that $z_i(c^e) \le (Q/\epsilon) - c^0$ when $c^e \in [0, r_i^e]$. The function $\Pi_i^*(z_i(c^e))$ is decreasing in $z_i(c^e)$ in the feasible region of $z_i(c^e)$. Moreover, since $z_i(c^e)$ is increasing in c^e, we know that $\Pi^*(z_i(c^e))$ is decreasing in c^e. ■

From Proposition 1, if one transportation mode has the lowest value of $z_i(c^e)$, i.e., the sum of carbon-emission tax $c^e e_i$, holding cost $\mathrm{hc}^0 l_i$, and transportation cost u_i, the alliance can maximize its profit by choosing that mode. The choice of transportation mode also influences the value of $z_i(c^e)$. Also, the government can influence the value of $z_i(c^e)$ by adjusting the carbon-emission tax c^e. It can be clearly seen that both transportation mode characteristics and carbon-emission tax are going to affect the optimal transportation mode selection. Next, we discuss how to choose the optimal transportation mode.

B. Optimal Transportation Mode Selection

Selecting an optimal transportation mode depends on $\Pi_1^*(z_1(c^e))$ and $\Pi_2^*(z_2(c^e))$. According to the discussion above, transportation mode k maximizes the alliance profit for $k = \mathrm{argmin}(z_i(c^e) : i \in \{1, 2\})$ and we can further get the optimal profit $\Pi_k^*(z_k(c^e))$. This transportation mode k will be used and is hence called the preferred transportation mode. The value of $z_i(c^e)$ is determined by the traditional logistics cost τ_i, emissions intensity e_i, and carbon-emission tax c^e. For smaller c^e, the cheaper and dirtier mode is better. For larger c^e, switching to the more expensive and less polluting mode is preferred. Next, we compare the sizes of $z_1(c^e)$ and $z_2(c^e)$. We define c_{12}^e such that $z_1(c_{12}^e) = z_2(c_{12}^e)$. Thus, we have

$$c_{12}^e = \frac{\tau_1 - \tau_2}{e_2 - e_1}. \quad (11)$$

Equation (11) represents an important threshold value of c^e. If $c^e < c_{12}^e$, then $z_1(c^e) < z_2(c^e)$. If $c^e > c_{12}^e$, then $z_1(c^e) > z_2(c^e)$. When $c_{12}^e \le 0$, i.e., $\tau_1 \ge \tau_2$, choosing transportation mode 2 can always generate more profit no matter what the value of c^e is. In this paper, the threshold c_{12}^e is also called the transportation mode shifting threshold (TMST). In addition, when the carbon-emission tax c^e is just equal to the threshold value c_{12}^e, we can get the following proposition.

Proposition 2: If $c^e = c_{12}^e$, then the equilibrium selling quantity, the equilibrium selling price, the maximum profit, and the consumer surplus obtained by choosing transportation mode 1 are the same as those obtained by choosing transportation mode 2. However, compared with transportation mode 1, choosing transportation mode 2 can reduce $q^* \times (e_1 - e_2)$ of carbon emissions, where $q^* = q_1(p_1^*(c_{12}^e)) = q_2(p_2^*(c_{12}^e))$.

Proof: From (8), if $c^e = c_{12}^e$, we have

$$q_1\left(p_1^*(c_{12}^e)\right) = \max\left\{\frac{1}{2}\left(Q - \epsilon \times \left(c^0 + z_1\left(c_{12}^e\right)\right)\right), 0\right\}$$

$$q_2\left(p_2^*(c_{12}^e)\right) = \max\left\{\frac{1}{2}\left(Q - \epsilon \times \left(c^0 + z_2\left(c_{12}^e\right)\right)\right), 0\right\}.$$

Hence, $q_1(p_1^*(c_{12}^e)) = q_2(p_2^*(c_{12}^e)) = q^*$. From (2), we have $\Gamma_1(p_1^*(c_{12}^e)) - \Gamma_2(p_2^*(c_{12}^e)) = e_1 \times q_1(p_1^*(c_{12}^e)) - e_2 \times q_2(p_2^*(c_{12}^e)) = q^* \times (e_1 - e_2)$. From (3), when $p_1^*(c_{12}^e) = p_2^*(c_{12}^e)$, we can show that $\mathrm{CS}_1(p_1^*(c_{12}^e)) = \mathrm{CS}_2(p_2^*(c_{12}^e))$. ■

We define a threshold value e_{12} as follows:

$$e_{12} = \left[e_1\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right)\right] \Big/ \left[\frac{Q}{\epsilon} - c^0 - \tau_1\right]. \quad (12)$$

From (12), we know that if $\tau_1 < \tau_2$, then $Q/\epsilon - c^0 - \tau_2 < Q/\epsilon - c^0 - \tau_1$ and $e_{12} < e_1$.

Proposition 3: When $\tau_1 < \tau_2$, the following results hold.

1) If $e_2 \le e_{12}$, then $r_2^e \ge r_1^e \ge c_{12}^e > 0$. Furthermore:
   a) $z_1(c^e) < z_2(c^e)$ for $c^e \in [0, c_{12}^e)$, i.e., $k = 1$ and $\Pi_1^*(z_1(c^e)) > \Pi_2^*(z_2(c^e))$;
   b) $z_1(c^e) \ge z_2(c^e)$ for $c^e \in [c_{12}^e, r_2^e)$, i.e., $k = 2$ and $\Pi_1^*(z_1(c^e)) \le \Pi_2^*(z_2(c^e))$;
   c) $\Pi_1^*(z_1(c^e)) = \Pi_2^*(z_2(c^e)) = 0$ for $c^e \in [r_2^e, +\infty)$.

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2) If $e_2 > e_{12}$, then $c_{12}^e > r_1^e > r_2^e > 0$. Moreover:
   a) $z_1(c^e) < z_2(c^e)$ for $c^e \in [0, r_1^e)$, i.e., $k = 1$ and $\Pi_1^*(z_1(c^e)) > \Pi_2^*(z_2(c^e))$;
   b) $\Pi_1^*(z_1(c^e)) = \Pi_2^*(z_2(c^e)) = 0$ for $c^e \in [r_1^e, +\infty)$.

Proof: In the following, we only show the detailed proof of Proposition 3.1 because the proof of Proposition 3.2 is similar to that of Proposition 3.1. First, define $\delta = e_{12} - e_2$, and we have $\delta \geq 0$. Substitute $e_2 = e_{12} - \delta$ into (9) yields

$$r_2^e = \frac{1}{e_{12} - \delta}\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right) \geq \frac{1}{e_{12}}\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right) = r_1^e. \tag{13}$$

Substitute $e_2 = e_{12} - \delta$ into (11) gives

$$\begin{aligned} c_{12}^e &= \frac{\tau_1 - \tau_2}{e_2 - e_1} = \frac{\tau_1 - \tau_2}{e_{12} - \delta - e_1} \leq \frac{\tau_1 - \tau_2}{e_{12} - e_1} \\ &= \frac{\tau_1 - \tau_2}{\frac{e_1\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right)}{\left(\frac{Q}{\epsilon} - c^0 - \tau_1\right)} - e_1} = r_1^e. \end{aligned} \tag{14}$$

From (13) and (14), we have $r_2^e \geq r_1^e \geq c_{12}^e > 0$. According to the definition of c_{12}^e, r_1^e, and r_2^e, we can show that $z_1(c^e) < z_2(c^e)$ for $c^e \in [0, c_{12}^e)$. From Proposition 1, we further have $\Pi_1^*(z_1(c^e)) > \Pi_2^*(z_2(c^e))$, $z_1(c^e) \geq z_2(c^e)$ for $c^e \in [c_{12}^e, r_2^e)$. For $c^e \in [r_2^e, +\infty)$, c^e is bigger than the BCRA, so we have $\Pi_1^*(z_1(c^e)) = \Pi_2^*(z_2(c^e)) = 0$. ∎

Since transportation mode 2 is more expensive, we show from Proposition 3 that the government could push the alliance to switch to mode 2 by imposing carbon-emission tax only if it is sufficiently clean (i.e., $e_2 \leq e_{12}$); otherwise, the alliance will keep choosing the cheaper mode 1 and react to the carbon-emission tax through increasing the selling price and decreasing the selling quantity. Moreover, under the condition of $\tau_1 \geq \tau_2$, transportation mode 2 will always be the preferred one. Moreover, under mode 2, the only way for the alliance to react to the carbon-emission tax regime is increasing the selling price and decreasing the selling quantity until getting out of the market (i.e., when $c^e > r_2^e$) or finding a mode cleaner than mode 2. According to the above analysis, whether the transport carbon-emission tax regime can encourage the alliance to shift to the cleaner transportation mode or not depends on: 1) the product characteristics; 2) the differences between different transportation modes; and 3) the market (including the maximum demand Q and the price-demand sensitivity coefficient ϵ).

C. Optimal Carbon Emission Tax

Substituting the mathematical expressions of the government revenues from carbon-emission tax, the social welfare losses due to carbon emissions, and the traditional welfare drivers of enterprise profits and consumer surplus into (6), we have

$$\begin{aligned} W(p_i, c^e) = &\left(p_i - \tau_i - c^0 - c^e e_i\right)(Q - \epsilon p_i) + c^e e_i (Q - \epsilon p_i) \\ &+ \frac{1}{2}\left(\frac{Q}{\epsilon} - p_i\right)(Q - \epsilon p_i) - c^s e_i (Q - \epsilon p_i). \end{aligned} \tag{15}$$

It is straight-forward to show that (15) can be expressed as follows

$$\begin{aligned} W(p_i, c^e) = &-\frac{1}{2\epsilon}(Q - \epsilon p_i)^2 + \frac{Q}{\epsilon}(Q - \epsilon p_i) \\ &- \left(\tau_i + c^0\right)(Q - \epsilon p_i) - c^s e_i (Q - \epsilon p_i). \end{aligned} \tag{16}$$

The main purpose of imposing carbon-emission tax on transport emissions is to decrease the social welfare losses due to carbon emissions by encouraging companies to reduce their volume of freight transport or select cleaner transportation modes. But at the same time, we can also observe from (16) that imposing the carbon-emission tax on transport emissions would decrease the sum of the government revenues and the traditional welfare drivers of company profits and consumer surplus. Therefore, the government needs to design an appropriate transportation carbon-emission tax to improve social welfare. Under the optimal production strategy of the alliance, the social welfare function is

$$\begin{aligned} W(p_i^*, c^e) = &-\frac{1}{8}\epsilon e_i^2 c^{e2} + \frac{1}{4}c^e \times \left(\epsilon e_i \times \left(\tau_i + c^0\right) - Q e_i + 2\epsilon e_i^2 c^s\right) \\ &+ \frac{3}{8}\epsilon \times \left(\tau_i + c^0 - \frac{Q}{\epsilon}\right)^2 + \frac{1}{2}c^s e_i \epsilon \times \left(\tau_i + c^0 - \frac{Q}{\epsilon}\right). \end{aligned} \tag{17}$$

Comparing the social welfare values added by the alliance's activities and the damages resulted from its negative externality, we can obtain the following proposition.

Proposition 4:
1) If $c^s \leq (3/4)r_i^e$, then $W(p_i^*, c^e = 0) \geq 0$.
2) If $c^s > (3/4)r_i^e$, then $W(p_i^*, c^e = 0) < 0$.

Proof: From (17), for $c^e = 0$, we have

$$\begin{aligned} W(p_i^*, c^e = 0) &= \frac{1}{2}\epsilon\left(\frac{Q}{\epsilon} - \tau_i - c^0\right)\left(\frac{3}{4e_i}\left(\frac{Q}{\epsilon} - \tau_i - c^0\right) - c^s\right) \\ &= \frac{1}{2}\epsilon\left(\frac{Q}{\epsilon} - \tau_i - c^0\right)\left(\frac{3}{4}r_i^e - c^s\right). \end{aligned} \tag{18}$$

Without the carbon-emission tax, the alliance can always get a positive profit no matter which transportation mode is used, so we have $((Q/\epsilon) - \tau_i - c^0) > 0$. We can further obtain $W(p_i^*, c^e = 0) \geq 0$ if $c^s \leq (3/4)r_i^e$ and $W(p_i^*, c^e = 0) < 0$ if $c^s < (3/4)r_i^e$. ∎

Proposition 4 shows that the social welfare value added by the company's business activities is smaller than the damages resulted from its negative externality if the value of SCC is analytically big enough. Next, we analyze how the intensity of transport carbon-emission tax can be set. We can see from (17) that $W(p_i^*, c^e)$ is concave in c^e. Let c_i^{e*} denote the extreme point of the function $W(p_i^*, c^e)$, then

$$c_i^{e*} = \frac{\tau_i + c^0 - \frac{Q}{\epsilon} + 2e_i c^s}{e_i}. \tag{19}$$

When $c_i^{e*} \in [0, r_i^e]$, c_i^{e*} is the optimal carbon-emission tax. In the following sections, we will analyze how the optimal carbon-emission tax can be set in different situations.

1) Optimal Carbon Emission Tax When No Existing Mode Shifts: When the product characteristics and the transport characteristics satisfy the conditions of Proposition 3.2 (i.e., $\tau_1 < \tau_2$ and $e_2 > e_{12}$), the alliance will

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keep selecting transportation mode 1 until the carbon-emission tax exceeds r_1^e. If the cleaner transportation mode 2 is also cheaper than transportation mode 1 (implying $\tau_2 \leq \tau_1$ and $e_2 < e_1$), transportation mode 2 will always be the most preferred mode. In the above situations, we can obtain the optimal carbon-emission tax through comparing c^s and r_i^e.

Proposition 5: For $i \in I$:

1) if $c^s \leq (1/2)r_i^e$, then $W(p_i^*, c^e)$ is decreasing in c^e for $c^e \in [0, r_i^e)$;
2) if $c^s \in ((1/2)r_i^e, r_i^e)$, then $W(p_i^*, c^e)$ is increasing in c^e for $c^e \in [0, c_i^{e*})$ and decreasing in c^e for $c^e \in [c_i^{e*}, r_i^e)$;
3) if $c^s \geq r_i^e$, then $W(p_i^*, c^e)$ is increasing in c^e for $c^e \in [0, r_i^e)$.

Proof: Without the carbon-emission tax, if we only choose transportation mode 1, the alliance can always get a positive profit. So we have $\tau_i + c^0 - (Q/\epsilon) > 0$. In order to further analyze whether imposing the carbon-emission tax can improve social welfare or not, we need to study the size relationship of c_i^{e*} and 0. According to (19), we can obtain

$$c_i^{e*} = 2c^s - \frac{\frac{Q}{\epsilon} - \tau_i - c^0}{e_i} = 2c^s - r_i^e. \tag{20}$$

It is easy to see from (20) that if $c^s \leq (1/2)r_i^e$, then $c_i^{e*} \leq 0$ and $c_i^{e*} - r_i^e = 2(c^s - r_i^e)$. Hence, if $c^s < r_i^e$, then $c_i^{e*} < r_i^e$. It is straight-forward to show that the function $W(p_1^*, c^e)$ is concave in c^e and the first-order condition yields $c^e = c_i^{e*}$. If $c^s \leq (1/2)r_i^e$, we know that $c_i^{e*} \leq 0 < r_i^e$ and $W(p_i^*, c^e)$ is decreasing in c^e for $c^e \in [0, r_i^e)$. If $c^s \in ((1/2)r_i^e, r_i^e)$, we know that $0 < c_i^{e*} < r_i^e$ and $W(p_i^*, c^e)$ is increasing in c^e for $c^e \in [0, c_i^{e*})$ and decreasing in c^e for $c^e \in [c_i^{e*}, r_i^e)$. If $c^s \geq r_i^e$, we know that $0 < r_i^e < c_i^{e*}$ and $W(p_i^*, c^e)$ is increasing in c^e for $c^e \in [0, r_i^e)$. ■

From Proposition 5, if: 1) the marginal benefit of reducing transport carbon emissions (i.e., SCC) is relatively small; 2) the emissions intensity is small; or 3) the economic activities can obtain a high yield (implying a high r_i^e), then imposing the carbon-emission tax may decrease social welfare. If c_i^{e*} is bigger than r_i^e, increasing the carbon-emission tax can always lead to social welfare improvement. However, whether the government should force the alliance to give up the business activities by making c^e larger than r_i^e or not requires the government to take into account some other factors, such as product substitutability, employment, and so on. From (9), we can see that r_i^e is decreasing in e_i, c^0, τ_i, and ϵ, and increasing in Q. Therefore, a bigger e_i, c^0, τ_i, or ϵ will lead to a higher probability of improving social welfare by imposing the carbon emissions tax. On the contrary, a bigger Q will yield a lower probability of improving social welfare by using the carbon emissions tax.

2) Optimal Carbon Emission Tax When Existing Mode Shifts: If the product characteristics and the transportation mode characteristics satisfy the conditions of Proposition 3.1, the competitiveness of transportation mode 2 will be higher when the carbon-emission tax increases. In this situation, the switch from modes 2 to 1 will occur (i.e., $\tau_1 < \tau_2$ and $e_2 < e_{12}$). We denote the difference between the social welfare values of choosing mode 1 and that of choosing mode 2 when $c^e = c_{12}^e$ by Δ_{12}

$$\Delta_{12} = W\big(p_2^*, c_{12}^e\big) - W\big(p_1^*, c_{12}^e\big).$$

It is easy to show that Δ_{12} can be rewritten as follows:

$$\Delta_{12} = \frac{1}{2}\epsilon\big(c^s - c_{12}^e\big)\left(e_1\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right) - e_2\left(\frac{Q}{\epsilon} - c^0 - \tau_1\right)\right). \tag{21}$$

Proposition 6: When the existing mode shifts:

1) if $c^s \geq c_{12}^e$, then $\Delta_{12} \geq 0$;
2) if $c^s < c_{12}^e$, then $\Delta_{12} < 0$.

Proof: From (21), we have $e_1((Q/\epsilon) - c^0 - \tau_2) - e_2((Q/\epsilon) - c^0 - \tau_1)$, which can be expressed as follows:

$$\begin{aligned} &e_1\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right) - e_2\left(\frac{Q}{\epsilon} - c^0 - \tau_1\right) \\ &\quad = e_1 e_2\left(\frac{1}{e_2}\left(\frac{Q}{\epsilon} - c^0 - \tau_2\right) - \frac{1}{e_1}\left(\frac{Q}{\epsilon} - c^0 - \tau_1\right)\right) \\ &\quad = e_1 e_2\big(r_2^e - r_1^e\big). \end{aligned} \tag{22}$$

From Proposition 3, if $\tau_1 < \tau_2$ and $e_2 < e_{12}$, we have $r_2^e > r_1^e$. Thus, (22) is always positive. Furthermore, we can complete the proof by noting that (21) is greater than or equal to 0 if $c^s \geq c_{12}^e$, and (21) is less than 0 if $c^s < c_{12}^e$. ■

Proposition 6 tells us that when the carbon-emission tax $c^e = c_{12}^e$, whether it can improve social welfare or not by switching from mode 1 to mode 2 depends on the size relationship of "transport shifting threshold" c_{12}^e and SCC (i.e., c^s). The reasons are as follows.

1) According to Proposition 2, when the carbon-emission tax $c^e = c_{12}^e$, the company profit and the consumer surplus of choosing mode 1 will be, respectively, equal to that of choosing mode 2. As a result, the value of Δ_{12} is determined by the difference of government revenues and the difference of social welfare losses resulting from the transport carbon emissions.
2) From Proposition 2, when the carbon-emission tax $c^e = c_{12}^e$, choosing mode 2 can reduce $q^* \times (e_1 - e_2)$ amount of carbon emissions compared to choosing mode 1, which implies that the government revenues will be decreased by $c_{12}^e q^* \times (e_1 - e_2)$ (a negative effect) and the "social welfare loss" resulting from transport carbon emissions will be decreased by $c^s q^* \times (e_1 - e_2)$ (a positive effect). According to the two reasons above, we find that when $c^e = c_{12}^e$, social welfare can be further improved by switching the transportation mode from mode 1 to mode 2 only when c^s is greater than the mode-shifting threshold c_{12}^e.

Next, we analyze how the carbon-emission tax influences the change tendency of welfare before and after the transportation mode shifting. It can be observed from (17) that the welfare function $W(p_i^*, c^e)$ is concave in the carbon-emission tax c^e. Thus, we can explore the change tendency of welfare by examining the size relationships of $c^s$and five thresholds c_{12}^e, r_2^e, $(1/2)r_1^e$, $(1/2)(r_1^e + c_{12}^e)$, and $(1/2)(r_2^e + c_{12}^e)$. Moreover, since $r_2^e \geq r_1^e \geq c_{12}^e > 0$ (from Proposition 3.1), the five thresholds themselves only have two types of size relationships (the settings 1 and 2 in Fig. 1). By analyzing the social welfare

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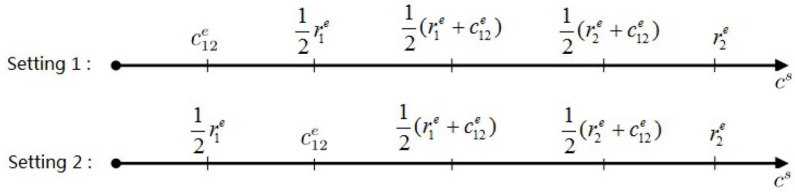


Fig. 1. Relationships of the thresholds.

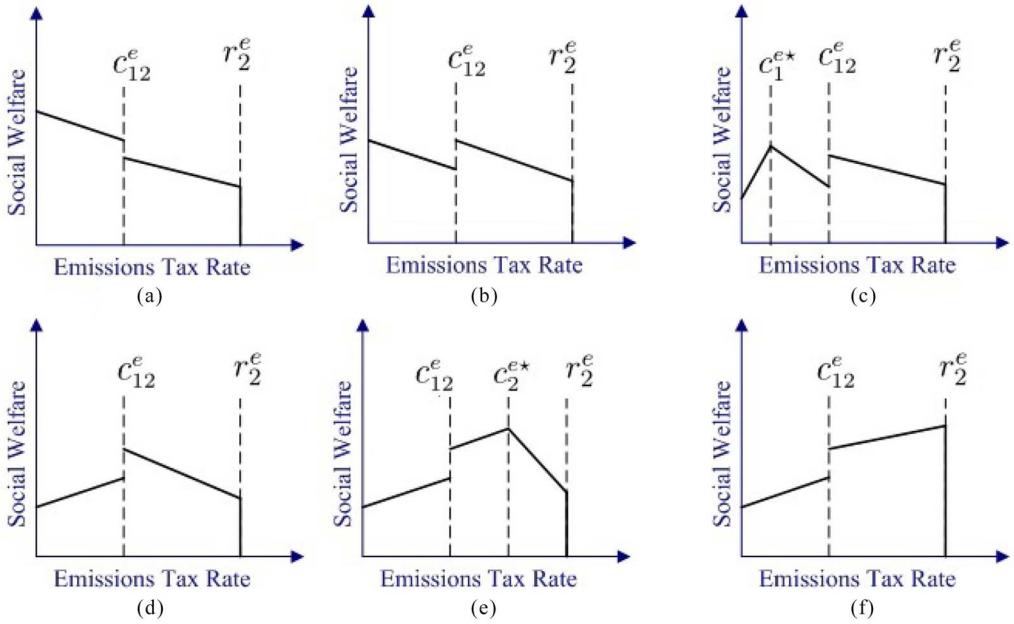


Fig. 2. Change tendency of social welfare. [(a)–(f) correspond to Proposition 7.1a–7.1f]

values before and after the transportation mode shifting when c^s lies in a different interval, we can derive Proposition 7 (see Fig. 2).

Proposition 7:

1) When $c^e_{12} < 1/2r^e_1 < 1/2(r^e_1+c^e_{12}) < 1/2(r^e_2+c^e_{12}) < r^e_2$, the following results hold.
   a) If $c^s \le c^e_{12}$, then $c^{e*} = 0$.
   b) If $c^e_{12} < c^s \le 1/2r^e_1$, then $c^{e*} \in \{0, c^e_{12}\}$.
   c) If $1/2r^e_1 < c^s \le 1/2(r^e_1+c^e_{12})$, then $c^{e*} \in \{c^{e*}_1, c^e_{12}\}$.
   d) If $1/2(r^e_1 + c^e_{12}) < c^s \le 1/2(r^e_2 + c^e_{12})$, then $c^{e*} = c^e_{12}$.
   e) If $1/2(r^e_2 + c^e_{12}) < c^s \le r^e_2$, then $c^{e*} \in \{c^e_{12}, c^{e*}_2\}$.
   f) If $r^e_2 < c^s$, then $c^{e*} \in \{c^e_{12}, r^e_2\}$.
2) When $1/2r^e_1 < c^e_{12} < 1/2(r^e_1+c^e_{12}) < 1/2(r^e_2+c^e_{12}) < r^e_2$, the following results hold.
   a) If $c^s \le 1/2r^e_1$, then $c^{e*} = 0$.
   b) If $1/2r^e_1 < c^s \le c^e_{12}$, then $c^{e*} = c^{e*}_1$.
   c) If $c^e_{12} < c^s \le 1/2(r^e_1 + c^e_{12})$, then $c^{e*} \in \{c^{e*}_1, c^e_{12}\}$.
   d) If $1/2(r^e_1 + c^e_{12}) < c^s \le 1/2(r^e_2 + c^e_{12})$, then $c^{e*} = c^e_{12}$.
   e) If $1/2(r^e_2 + c^e_{12}) < c^s \le r^e_2$, then $c^{e*} \in \{c^e_{12}, c^{e*}_2\}$.
   f) If $r^e_2 < c^s$, then $c^{e*} \in \{c^e_{12}, r^e_2\}$.

From Proposition 7, the optimal carbon-emission tax is influenced by: 1) SCC and 2) the factors that affect the five thresholds as shown in Fig. 1. The optimal carbon-emission tax depends on the relative size relationships among SCC and the five thresholds. The bigger value of SCC is, the higher the probability of improving social welfare becomes by imposing the carbon-emission tax. Note that, when SCC increases continuously, the optimal carbon-emission tax will not change continuously. Changes of the optimal carbon-emission tax occur only when SCC increases from one interval in setting 1 (setting 2) to another interval in setting 1 (setting 2).

Comparing Propositions 7.1 and 7.2, we find that different size relationships of the five thresholds will lead to different social welfare when the carbon-emission tax increases. Proposition 7 indicates that increasing SCC or decreasing any threshold in Fig. 1 can increase the probability of improving social welfare by imposing the carbon-emission tax. In addition, governments can always find an optimal c^e if $c^s \le r^e_2$. Increasing the value of c^e can always improve social welfare if $c^s > r^e_2$. From Proposition 7 and (11), we can obtain Corollary 1.

Corollary 1: When $\tau_1 < \tau_2$ and $e_2 < e_{12}$: 1) the smaller difference between τ_1 and τ_2 or the bigger difference between e_1 and e_2 is, the smaller value of c^e_{12} becomes and the higher probability of improving social welfare is by forcing companies to switch to the cleaner transportation mode 2; and 2) assume that the characteristic values of transportation modes 1 and 2 are fixed. The TMST c^e_{12} is increasing in the production cost c^0, decreasing in the volume v, and decreasing in the density ρ. Furthermore, the bigger v or ρ, or the smaller c^0 is, the higher the probability of improving social welfare becomes by forcing the companies to switch to the cleaner transportation mode 2.

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TABLE I
PROPERTIES OF TRANSPORTATION MODES

i	transportation cost c_i	emissions intensity e_i	lead time l_i
	$/km \bullet kg$	kg	day
Road	1.25×10^{-5}	$((2.089+0.3143d)10^{-4}v)\max(\rho,250)$	$\frac{d}{400}$
Rail	1×10^{-5}	$0.2223\times10^{-4}dv\rho$	$\frac{d}{240}$

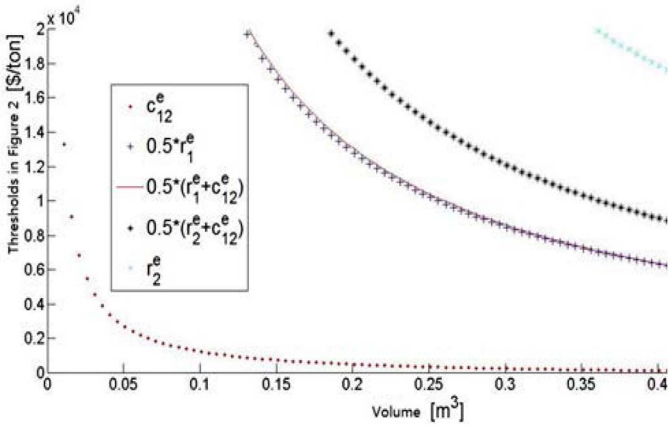


Fig. 3. Effects of v on the five thresholds.

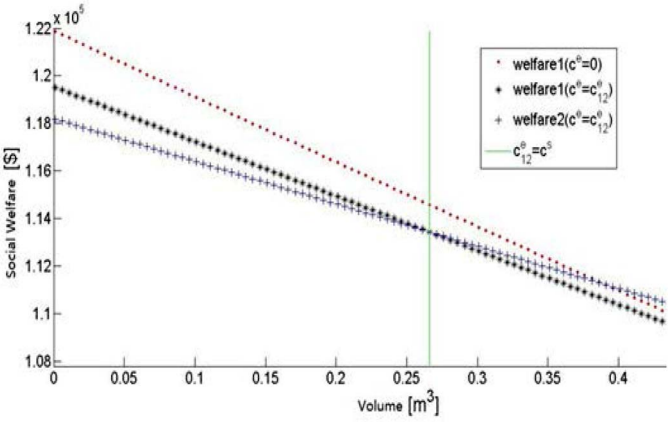


Fig. 4. Effects of v on social welfare.

IV. NUMERICAL EXAMPLES

This section presents the numerical analysis. According to the estimates for SCC [35], we set the value of SCC to be \$30–500/ton. We assume that the company will sell its product to a market from a factory 2000 km away, and the opportunity cost for the items in-transit is \$0.25/300/day, i.e., $h = 0.25/300$. The 3PL company provides two transportation modes: 1) mode 1 is the road transport and 2) mode 2 is the intermodal rail transport. Table I shows the transportation lead time, the transportation cost, and the emissions associated with transporting one unit of the product.

Example 1: We analyze how the product volume influences the critical thresholds and social welfare. In this example, we further specify the parameters as follows: $c^s = \$300$/ton, $c^0 = 1000\$$, $\rho = 1000$ kg/m^3, $Q = 4000$, and $\epsilon = 3$. We will observe how the above mentioned five thresholds (P.S.: shown in Fig. 1) change when v varies from 0.001 to 0.5 m^3. The results are shown in Fig. 3.

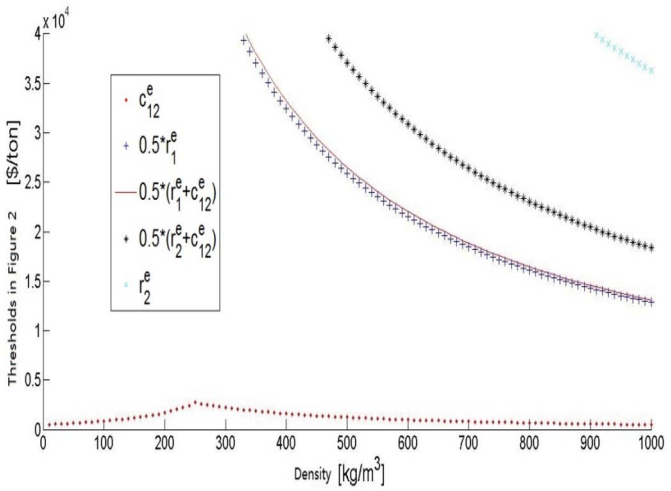


Fig. 5. Effects of ρ on the five thresholds.

Fig. 3 shows that all the five thresholds will decrease as v increases. With the exception of c^e_{12}, the chance of letting the other four thresholds to be below c^s is very small. This scenario is very common in practice because when c^0 is sufficiently big, the profit is usually much bigger than the carbon-emission tax, which leads to the results that both r^e_1 and r^e_2 will be substantially bigger than c^s. For social welfare, Fig. 4 reveals that: when $c^e = c^e_{12}$, if v is small and makes $c^s < c^e_{12}$, then $W(p^*_2, c^e_{12}) < W(p^*_1, c^e_{12})$; if v is larger and makes $c^s \geq c^e_{12}$, then forcing the company to choose the cleaner mode 2 will improve social welfare, i.e., $W(p^*_2, c^e_{12}) < W(p^*_1, c^e_{12})$. Compared with $W(p^*_1, c^e = 0)$, $W(p^*_2, c^e_{12})$ will be bigger only if $c^s - c^e_{12}$ is big enough (e.g., $v > 0.38$ in this example). In this situation, encouraging the company to choose mode 2 can improve social welfare.

Example 2: We analyze how the product density influences the five thresholds in Fig. 1. In this example, we specify the parameters as follows: $c^s = \$300$/ton, $c^0 = \$1000$, $v = 0.2$ m^3, $Q = 4000$, and $\epsilon = 3$. We will observe how the five threshold values change as ρ varies from 10 to 1000 kg/m^3. The results are shown in Fig. 5.

With the exception of c^e_{12}, the chance of having "the other four thresholds below c^s" is very small. The explanation is similar to that in Example 1. Also, we can see that the thresholds are decreasing in density. Note that the emissions and the transportation cost of road transport do not increase if the density is smaller than the prespecified density (i.e., 250 in this example). As a consequence, the rail transport becomes relatively more expensive if the density approaches 250. Thus, imposing the carbon-emission tax on the product with a sufficiently big or small density (with respect to 250 in this example) would have a high probability of improving social welfare.

Example 3: Let *Mp* denote "the ratio of Q and ϵ." We analyze the effects of c^0 and *Mp* on the values of c^e_{12}, $0.5\,r^e_1$, and $0.5r^e_2$. We discuss two scenarios in which the production cost is relatively low and the production cost is relatively

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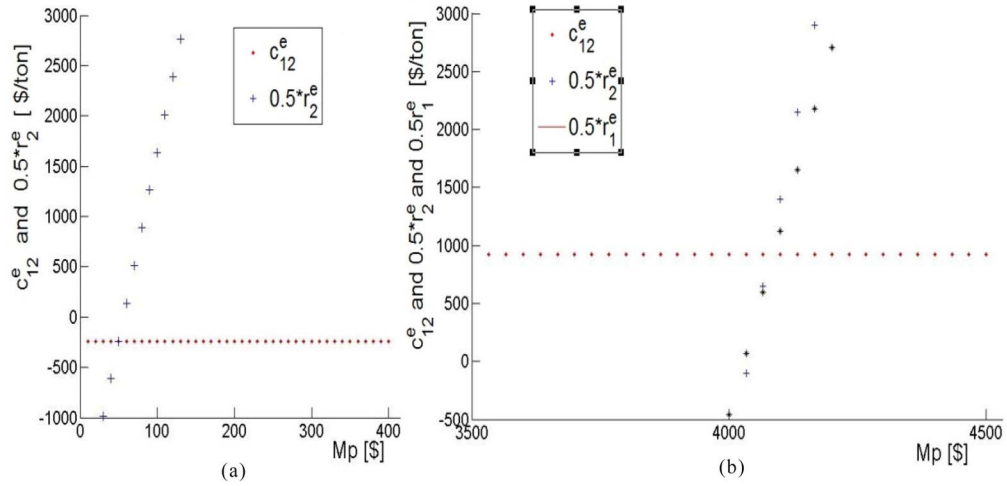


Fig. 6. Effects of *Mp* when the production cost is relatively (a) low and (b) high.

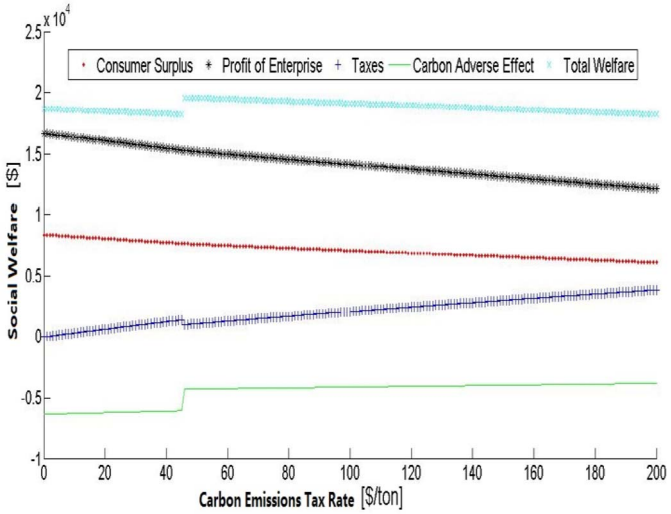


Fig. 7. Effects of carbon-emission tax.

high, respectively. In scenario 1, we specify the parameters as follows: $c^0 = \$50$, $v = 0.3$ m^3, $\rho = 1000$ kg/m^3, and $\epsilon = 3$, and we vary the value of *Mp* by changing Q from 0 to 1200 and the step is 30. The results are shown in Fig. 6(a). In scenario 2, we specify the parameters as follows: $c^0 = \$4000$, $v = 0.5$ m^3, $\rho = 1000$ kg/m^3, and $\epsilon = 3$, and we vary the value of *Mp* by changing Q from 1000 to 13 500 and the step size is 100. The results are shown in Fig. 6(b).

Fig. 6(a) shows that transportation mode 2 is the dominant mode in both transportation cost and transport carbon emissions since the holding cost is relatively small for the products with a low production cost. It can also be seen that $0.5r^e_2$ is increasing in *Mp*. Fig. 6(b) reveals that both $0.5r^e_1$ and $0.5r^e_2$ are increasing in *Mp*. However, c^e_{12} is fixed as *Mp* increases. The long lead time of transportation mode 2 leads to a very large holding cost under the high production cost case. Both $0.5r^e_1$ and $0.5r^e_2$ will be greater than 3000 when *Mp* increases to 110% of c^0. Actually, *Mp* is usually larger than 110% of c^0, and c^s is definitely smaller than 3000. Therefore, when either v or ρ is sufficiently large (resulting in a sufficiently small c^e_{12}) and the government makes $c^e = c^e_{12}$, the government can improve social welfare by imposing the carbon-emission tax.

Example 4: We analyze the changes of company's profit, consumer surplus, government revenues, welfare losses due to carbon emissions, and total social welfare. We specify the parameters as follows: $c^s = \$100$/ton, $c^0 = \$1000$, $v = 0.2$ m^3, $Q = 3500$, and $\epsilon = 3$. In particular, to make the trend more obvious, we increase the carbon emissions of road transport and rail transport to be ten times of the original values, and increase the transportation cost of road transport and rail transport to be five times of the original values. The results are shown in Fig. 7.

From Fig. 7, both the company's profit and the consumer surplus are decreasing in c^e, and the government revenue is increasing in c^e. However, the growth rate of government revenue is smaller than the reduction rate of company's profit. Therefore, increasing c^e will definitely decrease social welfare (without considering SCC). The only chance of improving social welfare by the carbon-emission tax regime comes from the reduction of social welfare loss resulting from carbon emissions. In this example, the value of emissions associated with transporting one unit of product is relatively small; decreasing social welfare losses resulting from carbon emissions by reducing the selling quantity cannot offset the loss to total social welfare. If $c^e = c^e_{12}$ and $c^s > c^e_{12}$, the total social welfare will have an improvement via shifting transportation mode from 1 to 2, because both the reduction of company's profit and the reduction of consumer surplus are zero, but the reduction of welfare loss resulting from carbon emissions (a positive effect) is larger than the reduction of government revenue (a negative effect).

V. Conclusion

Transportation mode selection is an important decision in industrial systems engineering. Different transportation modes have different carbon emission levels, and yield different social, environmental, and commercial effects. In this paper,

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we have analytically studied how carbon emissions affect the selection of transportation modes and social welfare.

In the conclusion, we summarize the core findings and insights as follows.

1) Our analytical results reveal that whether the transport carbon-emission tax can increase social welfare depends on the relationships among: 1) the SCC; 2) the TMST; and 3) the BCRA. Moreover, we know that a greater SCC, a smaller TMST, and a smaller BCRA would help lead to a higher probability of improving social welfare via imposing the transportation carbon emissions tax.
2) For the products with a low production cost, we have revealed that companies will tend to choose the slower while cheaper transportation mode which releases fewer emissions. Imposing the carbon-emission tax on these products' transportation systems will only make the companies reduce the selling quantity. If the products have a very large density or a very large volume (which makes the amount of emissions associated with transporting one unit of product very large), we have shown that the carbon-emission tax regime has the ability to improve social welfare (via imposing a reasonable carbon-emission tax on the products' emissions).
3) For the products with a high production cost, a bigger rate of carbon-emission tax would yield a larger competitive advantage to the transportation mode with a long lead time and high emissions. This makes this transportation mode the preferred one when the carbon-emission tax goes beyond TMST. The only opportunity to improve social welfare by imposing a carbon-emission tax on these products' transport emissions is that the products have a very high density or a very big volume, and the carbon-emission tax is equal to the value of TMST.
4) Our finding has indicated that it is wise for the government to impose an increasingly stringent (with a wider scope and a bigger intensity) carbon-emission tax regime on transport carbon emissions.

Similar to other modeling research, we admit that this paper has several limitations which open avenues for future research. For example, this paper assumes the presence of only two kinds of transportation modes. It will hence be interesting to consider a case where there are more than two transportation modes. Furthermore, future research can be conducted on the stochastic version of the problem and examine the impacts brought by different stochastic factors. It will also be interesting to expand the scope of study to cover some related problems, such as supply chain coordination [37], [41], [42], remanufacturing [38], carbon trade [39], and consumer returns [40].

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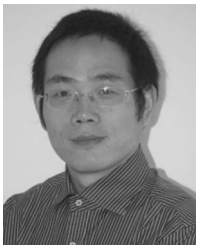
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